Seungwoo Eun

Contact Information

Busan, Republic of Korea
aglraswe@gmail.com
seungwooeun.github.io
LinkedIn
Google Scholar
ORCID 0009-0002-4848-9071

Research Interests

Spatial Computing, 3D User Interfaces, Immersive Visualization, Human-Computer Interaction, Extended Reality

Education

Pusan National University, Busan, Republic of Korea

M.S. in Artificial Intelligence GPA 4.38/4.5 — Mar. 2024 – Feb. 2026
Advisor Myungho Lee

Pusan National University, Busan, Republic of Korea

B.S. in Electrical and Computer Engineering GPA 3.25/4.5 — Mar. 2018 – Feb. 2024

Academic Experience

Pusan National University, XR Lab, Busan, Republic of Korea

Graduate Researcher Advisor Myungho Lee — Mar. 2024 – Feb. 2026

Spatial interaction across distance and scale. Investigated how spatial interfaces can support precise remote interaction across changes in distance and scale by designing and evaluating alternative portal-based techniques through a controlled user study.

Perception and movement in immersive environments. Examined how visual motion influences physical movement and perceptual stability during virtual locomotion through a controlled user study.

Undergraduate Research Intern — Mar. 2023 – Feb. 2024

Developed interactive communication systems with Unity and WebRTC and conducted computer vision research using SLAM.

Publications

Seungwoo Eun, Isaac Cho, Myungho Lee. Investigating Scale-Control Strategies for Portal-Based Remote Object Manipulation in Virtual Reality. In *IEEE International Symposium on Mixed and Augmented Reality*.
IEEE ISMAR 2026 · Conference Paper · To appear

Seungwoo Eun, Isaac Cho, Myungho Lee. Scalable Portals for Distant Object Manipulation in Virtual Reality. In *IEEE Conference on Virtual Reality and 3D User Interfaces*.
IEEE VR 2026 · Poster

Seungwoo Eun, Taeyun Noh, Myungho Lee. Effects of Peripheral Optic Flow Location and Speed on Unintended Positional Drift during Walk-In-Place in VR. In *IEEE International Symposium on Mixed and Augmented Reality*.
IEEE ISMAR 2025 · Conference Paper

Patents

Myeongho Lee, **Seungwoo Eun**, Point Cloud Skeletonization Method for 3D Reconstruction of Rod-shaped Structures.
Patent App. 10-2024-0171630 (KR), Pusan National University Industry-University Cooperation Foundation, 2024

Conference Presentations

Scalable Portals for Distant Object Manipulation in Virtual Reality.
Poster presentation, IEEE VR 2026, Mar. 2026.

Effects of Peripheral Optic Flow Location and Speed on Unintended Positional Drift during Walk-In-Place in VR.
Paper presentation, IEEE ISMAR 2025, Oct. 2025.

Invited Talks

The 17th Asia-Pacific Workshop on Mixed and Augmented Reality (APMAR) 2025 Sep. 2025
Pitch Your Work
❑ **Topic** Unintended Positional Drift during Walk-In-Place in VR

Teaching Experience

Pusan National University, Busan, Republic of Korea
Teaching Assistant, *Introduction to Data Science* Spring 2025
Supported an undergraduate course covering exploratory data analysis, statistical inference, k-means, k-nearest neighbors, and regression with Python.

Pusan National University, Busan, Republic of Korea
Teaching Assistant, *Adventure Design* Spring 2024
Supported an undergraduate course on IoT and embedded programming with Arduino, sensors, motor control, and team-based prototyping.

Academic Service

IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2026
Student Volunteer Oct. 2026

Honors and Awards

Third Place, Campus Capstone Design Competition, Pusan National University Nov. 2023
Real-time 3D reconstruction pipeline using NeRF

Skills

Development Unity, WebXR, ARKit, ARCore, ROS 2
Programming Languages Python, C++, C, C#, JavaScript
Languages Korean (Native), English (Near-native proficiency)